Course Code : ECT 259 **MQNR/RS – 23 / 1183**

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

PROBABILITY THEORY AND STOCHASTIC PROCESSES

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated against them.
(2) Assume suitable data wherever necessary.
(3) Due credit will be given to neatness.
(4) Use of Non – programmable calculator is permitted.
(5) Use of Standard Normal Distribution Table is permitted.

1. (a) A discrete random variable X has the following probability distribution :

x	0	1	2	3	4	5	6	7	8
f(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

Find :

(i) a,

(ii) $P(X < 3)$,

(iii) Mean of X. 3(CO1,5)

(b) Suppose that the duration in minutes of long distance telephone conversation follows an exponential density function PDF expressed as :

$$f(x) = 0{\cdot}2[e^{-x/5}], \quad \text{for} \quad x > 0$$

Determine the probability that duration of conversation :

(i) Will exceed **5** minutes.

(ii) Will be between **3** and **6** minutes.

(iii) Will be less than **3** minutes. 3(CO1,5)

(c) The chances of A, B and C becoming the General Manager of a company are in the ratio **4:2:3**. The probabilities that the bonus scheme will be introduced in the company if A, B and C become General Manager are 0·3, 0·7 and 0·8 respectively. If the bonus scheme has been introduced, what is the probability that A has been appointed as General Manager ? 4(CO1,5)

2. (a) Suppose that during a rainy season on a tropical island, the length of shower has an exponential distribution with average length of shower half a minute. What is the probability that a shower will last more than three minutes ? 3(CO1,5)

(b) Life Insurance salesman sales on the average 3 life insurance policies per week. Use Poisson's Law to calculate the probability that in a given week he will sale :

(a) 2 or more policies but less than 5 policies.

(b) Assuming that there are 5 working days per week, what is the probability that on a given day he will sale one policy. 3(CO1,5)

(c) What is the need of Moment Generating Function (MGF) of the random variables ?

Let the random variable Z is normally distributed $Z \sim N(0,\ 1)$. Find the MGF of random variable Z. 4(CO1,5)

3. (a) A company makes electronic gadgets. One out of every **50** gadgets is faulty, but the company doesn't know which ones are faulty until a buyer complains. Suppose the company makes a **$ 3** profit on the sale of any working gadget, but suffers a loss of **$ 80** for every faulty gadget because they have to repair the unit. Check whether the company can expect a profit in the long term. 3(CO1,3)

(b) Suppose that X is a random variable with mean and variance both equal to 16. Compute the **lower** bound on probability $P(0 < X < 32)$ using Chebyshev's inequality. 3(CO1,3)

(c) What do you understand from Markov's Inequality for Continuous and Discrete Random Variables ?

Suppose that number of customers visiting an ice-cream shop is a random variable with mean 40. Calculate the probability that number of customers visiting the shop will exceed 60. 4(CO1,3)

4. (a) A normally distributed IQ score have a mean of 100 and standard deviation of 15. Use the standard Z-table to answer following questions :

What is the probability of randomly selecting someone with an IQ score :

(a) Less than 80,

(b) Greater than 136,

(c) Between 95 and 110 ? 3(CO1,3)

(b) A certain group of welfare recipients receives SNAP benefits of $110 per week with a standard deviation of $20. If a random sample of 25 people is taken, Using CLT, find the probability that their mean benefit will be greater than $120 per week ? 3(CO1,3)

(c) In a communication system each data packet consists of 1000 numbers of bits. Due to the noise, each bit may be received in error with probability 0·1. It is assumed bit errors occur independently. Find the probability that there are more than 120 errors in a certain data packet.

(Let us define Xi as the indicator random variable for the i^{th} bit in the packet. That is, **Xi = 1** if the i^{th} bit is received in error, and **Xi = 0** otherwise. Then the Xi's are i.i.d. and Xi ~ Bernoulli (p = 0·1).) 4(CO1,3)

5. (a) Consider a Random Process $\{X(t),\ t \in R\}$ defined as

$X(t) = A \cos(w_0 t + \phi)$, where ϕ is Uniformly distributed i.e.

$\phi \sim U(0, 2\pi)$. **A** and w_0 are constants. Compute **Mean of X(t)**.
3(CO2,4)

(b) What do you understand by Random Processes ? Give classification of Random processes. 3(CO2,4)

(c) Given a random process

$X(t) = A \cos[2\pi f_c t + \theta]$, where θ is a random variable with uniform probability distribution in the interval $[0, 2\pi]$. Check whether the given Random process is Wide sense stationary. 4(CO2,4)

6. (a) A random process $\mathbf{X(t)}$ having auto-correlation function :

$\mathbf{R_{XX}(\tau) = e^{-4|\tau|}}$ is applied as input to the **LTI system** with impulse response $\mathbf{h(t) = e^{-2t}u(t)}$.

Find the **PSD** $\mathbf{S_{YY}(\omega)}$ of output $\mathbf{Y(t)}$. 3(CO2,4)

(b) Find power spectral density of random process :

$\mathbf{X(t) = 10 \cos [2000\pi t + \theta]}$ where θ is a random variable with a uniform PDF in the interval $[-\pi, \pi]$. 3(CO2,4)

(c) Explain the processing of random signals through an LTI system. 4(CO2,4)